

Summary - 2403.06535V1 Copy

Original File: 2403.06535v1_copy.pdf | Generated on: June 13, 2026

1. Overall Summary

This paper introduces DeLAMA, a novel decentralized and lifelong-adaptive multi-agent collaborative learning framework designed to overcome the limitations of centralized and static collaboration methods in multi-agent systems. The core problem addressed is enabling agents to autonomously identify beneficial collaborators and adapt to evolving tasks without relying on a central server or predefined relationships. DeLAMA achieves this through a decentralized graph structure learning algorithm for dynamic collaboration graphs, a memory unit to capture accumulated learning history while maintaining finite storage, and algorithm unrolling that transforms iterative optimization into a neural network. This hybrid approach allows agents to "learn to collaborate" through supervised training. Theoretical analysis confirms communication efficiency and convergence properties. Experimental results across regression, image classification, and multi-robot mapping tasks demonstrate DeLAMA's superior performance, achieving up to a 98.80% reduction in MSE and a 188.87% improvement in classification accuracy compared to existing methods, while also showing more effective collaboration discovery and adaptation to dynamic scenarios.

2. Main Idea

The central concept is a novel decentralized and lifelong-adaptive multi-agent collaborative learning framework, named DeLAMA, which enables multiple agents to autonomously identify beneficial collaboration relationships and adapt to dynamically changing tasks without a central server. It combines mathematical optimization and neural networks through algorithm unrolling to allow agents to "learn to collaborate" efficiently and robustly in dynamic environments.

3. Problem Being Solved

This paper addresses two major weaknesses in previous multi-agent collaborative learning systems: 1) **Reliance on a central server:** Many existing methods (e.g., federated learning, multi-task learning) use a central server to coordinate collaboration, which introduces vulnerability (central server failure disables all collaboration) and communication overhead. 2) **Static collaboration relationships:** Previous approaches often involve predefined, fixed collaboration relationships (e.g., fully connected graphs, star graphs) or relationships calculated by a central server. These static structures are inadequate and inflexible for dynamic scenarios where agents' tasks and potential relationships evolve over time, limiting collaboration efficiency.

4. Assumptions

- The training data arrives independently among agents at each time t , and data batches satisfy conditional independency on both agent-level and time-level.
- Similar model parameters promote collaboration, meaning stronger collaboration relations correspond to more similar model parameters, guiding the prior distribution of model parameters.
- The adjacency matrix $W(t)$ is symmetric, which is assumed for rewriting the linear equation in the lifelong model update.
- For the theoretical analysis of the best point for Taylor expansion (Theorem 1), the model is assumed to be linear and the loss function is a standard supervised training loss (e.g., mean square error or cross-entropy).
- For classification tasks, the Hessian matrix is assumed to be Lipschitz continuous.
- The matrix $B(t)$ is invertible, which is assumed for the parameter updating rule in the lifelong model update.

5. Limitations

- The iterative algorithm (before algorithm unrolling) has potential approximation errors for non-linear functions due to Taylor expansion, limiting its expressive capability for non-linear learning tasks.
- The iterative algorithm involves many hyperparameters, demanding significant effort for searching and tuning.
- The iterative algorithm requires numerous iterations (for graph learning, message passing, and alternating convex search), leading to significant communication overhead and making it less practical for real applications.

6. Results and Conclusion

DeLAMA consistently outperforms existing federated learning, lifelong learning, and decentralized optimization methods across various tasks: 1) **Regression:** Achieved a 98.80% reduction in Mean Square Error (MSE) at the last timestamp (t=10) compared to baselines. It effectively discovers collaboration strategies and adapts to dynamic learning scenarios, learning smooth and accurate collaboration graphs similar to the oracle. 2) **Image Classification (MNIST & CIFAR-10):** Demonstrated a 99.66% and 134.01% improvement in accuracy on MNIST and CIFAR-10 respectively at t=10 compared to classic lifelong learning. It also showed a 54.42% (MNIST) and 188.87% (CIFAR-10) improvement in classification accuracy compared to methods using static collaboration graphs. DeLAMA accurately learns evolving collaboration graphs, adapting to agents' observations. 3) **Multi-Robot Mapping:** Achieved the highest F1-score (95.31% at t=12) compared to other decentralized collaborative model training approaches, significantly improving environmental exploration efficiency and performance, and producing more complete room structures. 4) **Human-Involved Experiment:** DeLAMA consistently achieved lower regression MSE than human participants across different data qualities. The human collaboration mechanism showed a negative correlation between model parameter distance and collaboration strength, similar to DeLAMA's underlying assumption. In conclusion, DeLAMA effectively enables decentralized, lifelong-adaptive collaboration, dynamically infers collaboration relationships, and adapts to evolving tasks, demonstrating superior performance and robustness.

7. Real World Impact

The research provides a foundational technique for developing intelligent, decentralized, and dynamic multi-agent systems. Practical applications include: 1) **Autonomous driving (V2X communication):** Agents (vehicles) can collaborate to share information and make decisions without a central authority, adapting to changing road conditions and other vehicles. 2) **Multi-robot environment exploration/mapping:** Robots can collaboratively map unknown environments, sharing information and adapting their strategies dynamically to achieve a more complete and efficient map. 3) **Edge AI/Distributed Machine Learning:** Enables collaborative training of machine learning models across multiple clients or edge devices without a central server, enhancing robustness and efficiency. 4) **Smart healthcare/IoT:** Decentralized collaboration among sensors and devices for data analysis and decision-making, adapting to dynamic patient conditions or environmental changes.

8. Graphs and Figures Analysis

Figure 1: The illustration of our decentralized and lifelong-adaptive multi-agent collaborative learning system.

This figure visually represents the core concept of DeLAMA. It shows six agents (represented by owls) collaborating over three timestamps (Timestamp 1, 2, 3) in a dynamic environment. The agents have different observations (indicated by varying colors around them) and their model parameters (small circles) are updated. Crucially, the collaboration relationships (lines connecting agents) dynamically adapt over time, reflecting the lifelong-adaptive nature of the system. The legend clarifies the symbols: model parameters, collaborating agents, different observations, and collaboration relationships. This figure highlights the system's ability to autonomously identify and adapt collaboration partners in a decentralized manner.

Figure 2: Relationships among our lifelong collaborative learning and its related works: lifelong learning, federated learning, and multi-task relation learning.

This 2x2 matrix categorizes different collaborative learning paradigms based on two dimensions: "Dynamic scenario" vs "Static scenario" (x-axis) and "Predefined collaboration relations" vs "Learned collaboration relations" (y-axis). Lifelong learning (top-left) is dynamic but uses predefined relations. Federated learning (bottom-left) is static and uses predefined relations. Multi-task relation learning (bottom-right) is static but learns relations. DeLAMA, labeled "Decentralized and lifelong-adaptive collaborative learning" (top-right), uniquely occupies the quadrant of dynamic scenarios with learned collaboration relations. This figure effectively positions DeLAMA within the existing research landscape, emphasizing its novel combination of features.

TABLE 1: Relations to previous task settings, including lifelong learning, multi-task learning, and federated learning.

This table provides a concise comparison of DeLAMA with other learning settings based on three criteria: "Dynamic scenario", "Decentralized mechanism", and "Number of agents". Lifelong learning supports dynamic scenarios but is centralized and typically for a single agent. Multi-task learning and Federated learning are neither dynamic nor decentralized, but involve N agents. DeLAMA (Ours) is the only approach that simultaneously supports dynamic scenarios (✓), a decentralized mechanism (✓), and involves multiple (N) agents. This table clearly highlights DeLAMA's unique capabilities in addressing the limitations of previous methods.

Figure 3: The decentralized and lifelong-adaptive multi-agent collaborative learning system.

This figure illustrates the three main steps of the DeLAMA system: Local learning, Collaborative relational inference, and Lifelong model update. Local learning involves agents learning model parameters from their own observations. Collaborative relational inference involves agents transmitting parameters and learning collaboration relationships. Lifelong model update involves agents sharing parameters based on learned relationships and refining their models. The figure shows that the "Collaboration relational inference" and "Lifelong model update" steps iterate until convergence, indicating a feedback loop. The legend clarifies the symbols for model parameters, collaborating agents, different observations, collaboration relationships, and communication graph constraints. This diagram provides a detailed workflow of how DeLAMA operates.

Figure 4: The unrolled network structure of the collaboration system with input data at one exact time.

This figure depicts the neural network architecture resulting from algorithm unrolling in DeLAMA. It shows how the iterative optimization steps are transformed into layers of a deep neural network. The inputs are training data, learning history (memory), and the communication graph. The Local Learning layer processes input data through a backbone network and Taylor expansion to calculate initial model parameters. Iterative layers (Layer 1 to Layer M) represent the unrolled iterations of collaborative relational inference and lifelong model update, where agents communicate, infer collaboration relationships, and update model parameters. The output is the final model parameters. The network parameters are learnable, enabling the system to learn optimal collaboration strategies. This figure illustrates how DeLAMA integrates mathematical optimization with neural networks for enhanced learning and efficiency.

TABLE 2: Performance comparison for the regression problem.

This table presents a quantitative comparison of DeLAMA's performance (MSE at $t=1$ and $t=10$) on a regression task against various federated learning and decentralized optimization methods, as well as ablation studies (DeLAMA-WC: without collaboration, DeLAMA-WM: without lifelong-adaptive learning). DeLAMA consistently achieves the lowest MSE at both $t=1$ (2.3377) and $t=10$ (0.0719), significantly outperforming all other methods. The ablation studies show that both collaboration (DeLAMA-WC's higher MSE) and lifelong-adaptive learning (DeLAMA-WM's higher MSE) are crucial components for DeLAMA's superior performance. This table demonstrates DeLAMA's effectiveness and the importance of its core mechanisms.

TABLE 3: The evaluation of the collaboration performance compared with other centralized collaboration mechanisms.

This table evaluates DeLAMA's collaboration performance (GMSE and MSE at $t=1$, $t=5$, $t=10$) on the regression problem under different communication graph constraints (Fully Connected (FC), Erdos-Renyi (ER), Barabasi Albert (BA)). It compares DeLAMA with other relational learning methods (NS, GLasso, MTRL, GL-LogDet, L2G-PDS) and an Oracle baseline. DeLAMA consistently achieves low GMSE and MSE across all communication structures, often outperforming or matching the Oracle baseline and significantly surpassing other methods. The results show that DeLAMA effectively learns collaboration relationships and maintains strong task performance even under varying communication constraints and low connectivity levels.

Figure 5: Visualization of the learned graph structure at time $t = 3$ compared with the oracle collaboration structure.

This figure displays heatmaps representing the learned collaboration graph structures (adjacency matrices) at time $t=3$ for various methods (GL-LogDet, GLasso, MTRL, L2G-PDS, NS, DeLAMA (Ours)) and the Oracle ground truth. The heatmaps show the strength of collaboration between agents (rows/columns 0-5), with darker squares indicating stronger collaboration. DeLAMA's learned graph structure closely resembles the Oracle graph, exhibiting smooth and accurate edge weights. In contrast, other baseline methods show less accurate or sparser collaboration patterns, failing to capture the true relationships as effectively. This visualization qualitatively demonstrates DeLAMA's ability to accurately infer collaboration relationships.

Figure 6: A collaboration result of two groups of agents that are performing two different regression tasks.

This figure illustrates DeLAMA's dynamic learning and collaboration capabilities in a regression task involving two groups of agents. Subfigure (a) shows the regression lines learned by individual agents over time ($T=1$ to $T=5$). Initially, agents' learned functions might be inaccurate, but as time progresses and they collaborate, their learned functions converge closer to the true underlying functions, demonstrating learning from past data. Subfigure (b) displays heatmaps of the collaboration graph at different timestamps ($T=1$ to $T=5$). The heatmaps show that the collaboration relationships dynamically evolve, forming two distinct groups of agents that collaborate internally, reflecting the two different regression tasks. This visualization highlights DeLAMA's ability to adapt collaboration structures over time.

TABLE 4: The system performance analysis on classification tasks of four different settings.

This table presents the classification accuracy (Acc at $t=1$, $t=5$, $t=10$) of DeLAMA compared to various lifelong learning, federated learning, and decentralized optimization methods on MNIST and CIFAR-10 datasets. It also includes ablation studies (DeLAMA-WC: without collaboration, DeLAMA-WM: without lifelong-adaptive learning). DeLAMA achieves the highest accuracy at $t=10$ for both MNIST (99.51%) and CIFAR-10 (76.03%), significantly outperforming all other baselines. The ablation studies show that both collaboration and lifelong-adaptive learning are critical for DeLAMA's superior classification performance. This table quantitatively demonstrates DeLAMA's effectiveness in classification tasks and its ability to remember previously learned data and adapt to new knowledge.

TABLE 5: The system collaborative learning performance of different collaboration relationship learning methods.

This table compares DeLAMA's performance (GMSE and classification accuracy at $t=1$, $t=3$, $t=5$) against other collaboration relationship learning methods (NS, GLasso, MTRL, GL-LogDet, L2G-PDS) and an Oracle baseline on MNIST and CIFAR-10 datasets. DeLAMA consistently achieves the lowest GMSE and highest accuracy across both datasets and timestamps, often closely matching or surpassing the Oracle baseline. This table reinforces DeLAMA's superior ability to learn accurate collaboration graphs and leverage them for improved task performance in classification scenarios, outperforming traditional and other structural learning methods.

Figure 7: Example of the evolving collaboration graphs.

This figure visualizes the time-evolving collaboration graphs (heatmaps) produced by DeLAMA for MNIST (a) and CIFAR-10 (b) datasets at different timestamps ($T=1, 3, 5, 7$), compared to the Oracle ground truth. The heatmaps show that DeLAMA's learned collaboration relationships dynamically adapt over time. Initially, agents might collaborate more broadly ($T=1$), but as their observations grow, the system tends to prune less useful relationships and strengthen beneficial ones, forming clearer collaborative groups ($T=3, 5, 7$). The evolving patterns closely resemble the Oracle structure, demonstrating DeLAMA's ability to dynamically infer and refine collaboration relationships in response to changing task observations.

TABLE 6: The agents' average F1-score of the multi-robot mapping task.

This table presents the average F1-score (at $t=1$, $t=6$, $t=12$) of DeLAMA compared to other decentralized collaborative model training approaches (DSGD, DSGT, DiNNO) on a multi-robot mapping task. It also includes an ablation study (DeLAMA-WC: without collaboration). DeLAMA achieves the highest F1-score at $t=12$ (95.31%), significantly outperforming all other methods. The comparison with DeLAMA-WC (68.50% at $t=12$) clearly shows that multi-agent collaboration is crucial for improving mapping performance. This table quantitatively demonstrates DeLAMA's effectiveness in real-world multi-robot collaboration scenarios.

Figure 8: The simulation result of robot mapping at $t = 12$ based on the CubiCasa5k room Dataset.

This figure shows visual comparisons of the mapped room structures at $t=12$ for the Truth Map, DeLAMA, DeLAMA-WC (single agent mapping), DSGD, DSGT, and DINNO. Black represents empty space, and white represents occupied space. DeLAMA's mapped room (second image) is visually the most complete and accurate, closely resembling the Truth Map. DeLAMA-WC (without collaboration) shows a much sparser and incomplete map, highlighting the benefit of collaboration. Other decentralized methods (DSGD, DSGT, DINNO) also produce less complete maps compared to DeLAMA. This figure qualitatively demonstrates DeLAMA's superior performance in multi-robot mapping, enabling a more comprehensive understanding of the environment through collaboration.

TABLE 7: The average performance of human collaboration compared with DeLAMA.

This table compares the average regression Mean Square Error (MSE at $t=1$, $t=2$, $t=3$) between human participants and DeLAMA under different data quality levels (Low, Medium, High) in a human-involved experiment. DeLAMA consistently achieves lower MSE than human collaboration across all data quality levels and timestamps. For example, at $t=3$ with high data quality, DeLAMA's MSE is 0.0112 compared to human's 0.0270. This table quantitatively shows that DeLAMA's automated collaboration strategy is more effective than human-driven collaboration in this specific cognitive learning task.

Figure 9: The human evaluation of collaboration relationships.

This figure presents scatter plots showing the relationship between the logarithm of model parameter distance ($\log(D_{ij})$, y-axis) and human rating scores for collaboration strength (x-axis, 0=lowest, 5=highest) under different data quality levels (High, Medium, Low). The plots show a general negative correlation: as human collaboration scores increase (higher collaboration strength), the model parameter distance between agents tends to decrease. This verifies the paper's assumption that humans tend to collaborate with similar knowledge. The distribution of points suggests that for lower data quality (rightmost plot), humans find it harder to distinguish useful collaborators, indicating more complex underlying mechanisms beyond simple similarity. This figure provides qualitative insights into human collaboration behavior and its alignment with DeLAMA's design principles.

Figure 10: A time-evolving learning task example of two agents sampled from the regression task dataset.

This figure illustrates the progression of a regression learning task for two agents over five timestamps ($T=1$ to $T=5$), along with the ground truth function. Each subplot shows scatter points (training samples) and the learned regression line for each agent. Initially, at $T=1$, agents have limited samples and their learned functions are far from the ground truth. As time progresses ($T=2$ to $T=5$), agents accumulate more samples (indicated by wider sampling intervals) and their learned regression lines converge closer to the ground truth function. The figure demonstrates how agents' knowledge of the target function improves over time through lifelong learning, even with limited initial observations.

Figure 11: The human-machine interaction framework used for the human-involved experiment.

This diagram illustrates the hierarchical framework designed for the human-involved experiment. It shows human-machine interaction where humans interact visually and provide input, while virtual agents imitate human behaviors. In the background, the core DeLAMA algorithm runs, with "proxy machines" facilitating interaction between humans and the DeLAMA agents. The framework enables humans to observe others' behaviors, submit their understandings, and allows virtual agents to perform visual interaction. This top-down design provides a platform for real human evaluation of collaborative learning.

Figure 12: The login page of the human-involved experiment, contains a user guide for the participants to learn how to play the game.

This screenshot shows the login interface for the human-involved experiment. It includes fields for User ID and Username, and a "How to play the game?" section that provides instructions for participants on the linear regression task, emphasizing the challenge of partial data and the opportunity to modify answers based on others' results. The small graphs at the top right show example learning progress over time. This figure illustrates the initial entry point for participants in the experiment.

Figure 13: The local learning page of the human-involved experiment. Here participant's drawn line will be transformed into a linear equation with two parameters.

This screenshot displays the local learning interface for a human participant in the experiment. It shows a scatter plot of data points and a drawn line representing the participant's current guess for the regression line. The equation of the drawn line ($y = 0.39x + -4.60$) and its parameters (K and b) are displayed. The "Submit" button allows the participant to finalize their local learning result. This page represents the initial phase where each agent (human) forms their understanding based on their individual observations.

Figure 14: The aggregation page of the human-involved experiment, which contains other participants' results and the score rated by each agent.

This screenshot shows the "Result Modification" page, where a human participant can view the learning results (regression lines) of other participants. The participant's own line is also shown, and they can modify their initial learned regression results based on the collective information. The page also displays an "Evaluation of your performance" score and allows for submission. This page represents the collaborative phase where agents share information and refine their models.

Figure 15: The final page of the human-involved experiment with a leaderboard, each participant can look back to historical learning results.

This screenshot displays the "Game finished" page, which includes a leaderboard showing the average performance of participants and a "Result Log" graph. The graph visualizes the historical learning results (evaluation loss over time) for different participants, along with the ground truth regression target function. This page allows participants to review their performance and the overall game outcome, providing a summary of the experiment.